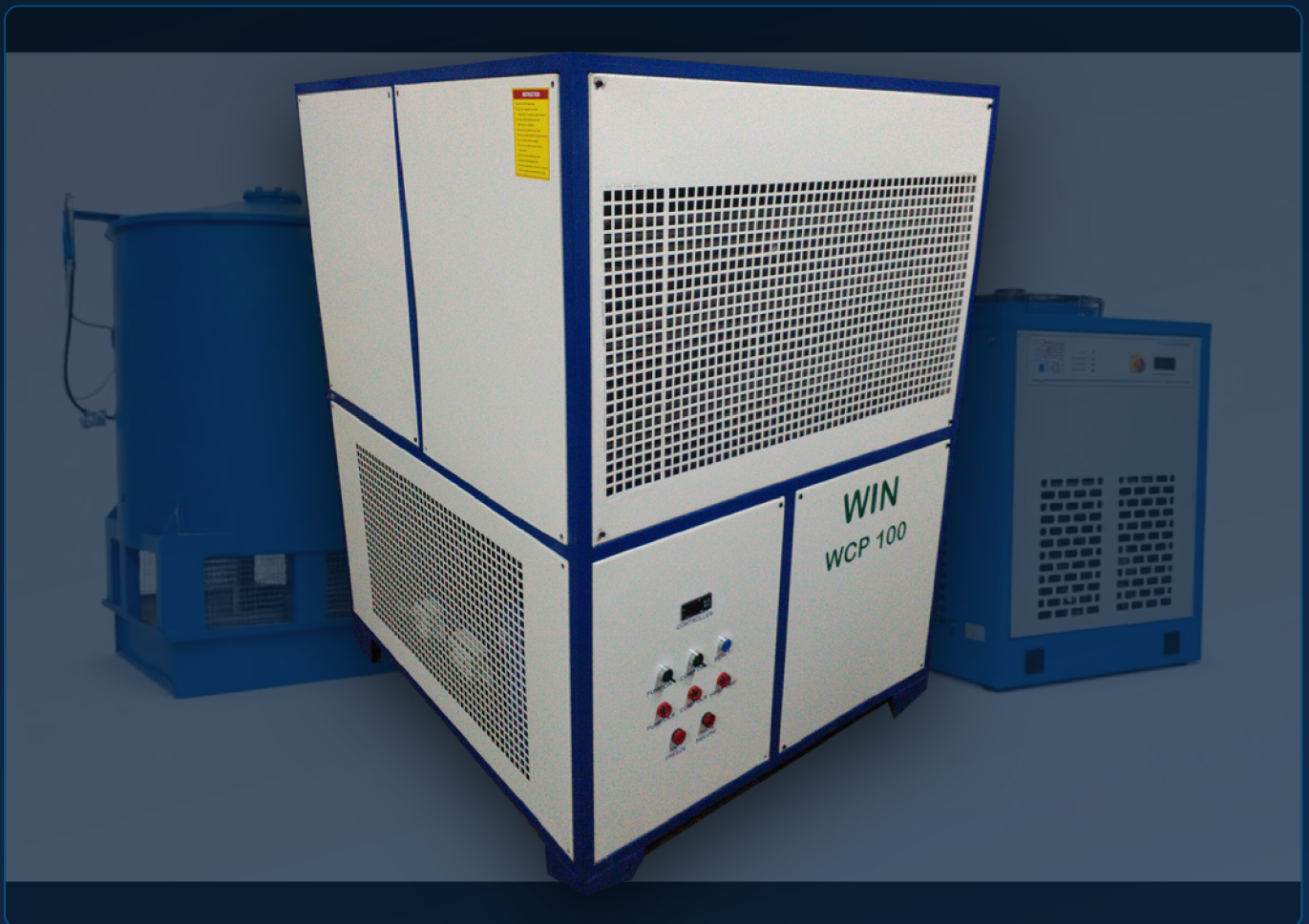


WCP SERIES (0.5 TO 50 TR)

High-Precision Industrial Process Chillers

Packaged Closed-Loop Air & Water-Cooled Chillers with Integrated SS304 Buffer Reservoir & Stainless Centrifugal Circulation Pumps



±0.5°C STABILITY

Digital Microprocessor
PID Temperature Control

SS304 BUFFER TANK

15L to 700L Heavy
Reservoir Dampens
Spikes

SCROLL COMPRESSORS

Copeland & Danfoss
High-COP Technology

NON-FERROUS LOOP

100% Stainless Wetted
Fluid Path Prevents Rust

THERMODYNAMIC ARCHITECTURE & ENGINEERING

Operational Overview & Engineering Advantages

Continuous industrial duty engineered for 45°C tropical ambient conditions

Win Equipments WCP Series industrial process chillers provide closed-loop liquid thermal management for factory automation, plastic molding, laser cutting, and chemical processing. Built to withstand continuous 24/7 industrial duty in high-ambient environments up to 45°C, WCP chillers integrate high-efficiency scroll compressors, a heavy-gauge insulated SS304 buffer reservoir, and an all-stainless centrifugal circulating pump within a compact powder-coated steel monoblock skid.

Working Principle: Process fluid absorbs heat from industrial machinery and returns to the internal SS304 insulated buffer tank. A stainless centrifugal pump draws fluid from the tank and circulates it through a 316 stainless steel brazed plate heat exchanger (or shell-and-tube evaporator) where heat is transferred to eco-friendly R410A/R407C refrigerant. The cooled fluid is delivered back to process equipment at user setpoints from +5°C to +25°C with $\pm 0.5^\circ\text{C}$ PID stability. The large thermal mass of the buffer tank dampens sudden thermal shocks caused by cyclic machine operations.

SYSTEM INTEGRATION & P&ID FLOW ARCHITECTURE

Machinery Thermal Load → Return Header → SS304 Buffer Tank → SS Centrifugal Pump → SS316 Evaporator Plate → Modulated Refrigeration Loop → Precision Chilled Supply Header → Machine Tools

All-Stainless SS304 Non-Ferrous Wetted Loop

Internal buffer tank, pump impellers, and brazed plate heat exchanger are manufactured from stainless steel. Prevents galvanic corrosion, rust slurry, and clogged cooling channels in expensive molds and laser optics.

High-Efficiency Copeland / Danfoss Scroll Compressors

Utilizes hermetic scroll compressors with 70% fewer moving parts than reciprocating machines. Delivers industry-leading COP exceeding 4.2, quiet operation, and exceptional tolerance against liquid refrigerant slugging.

Modulating Hot Gas Bypass Valve

Mechanical hot gas bypass system modulates refrigeration capacity down to 10% thermal load. Maintains tight $\pm 0.5^\circ\text{C}$ temperature stability without compressor cycling during machine idling or mold changeovers.

Intelligent Microprocessor & Safety Interlocks

Central control panel provides digital setpoint programming, RS-485 Modbus telemetry, low fluid level protection, anti-freeze thermostat, and reverse-phase compressor protection.



CERTIFIED ENGINEERING RATINGS

Technical Specifications Matrix

Full standard model range & operational parameters

Standard Design Conditions: Leaving Water Temp: +12°C (Range +5°C to +25°C) | Water Delta T: 5°C | Ambient Temp: 35°C (Rated up to 45°C) | Refrigerant: R410A / R407C Eco-friendly | Buffer Tank: SS304 Insulated

MODEL CODE	NOMINAL TR	COOLING KW	COMP POWER KW	SUPPLY VOLTAGE	WATER FLOW LPM	PUMP HP	TANK LITERS	CONNECTION SIZE
WCP 005	0.5 TR	1.7 kW	0.9 kW	230V / 1Ph / 50Hz	6 LPM	0.5 HP	15 L	3/4" BSP
WCP 010	1.0 TR	3.5 kW	1.5 kW	230V / 1Ph / 50Hz	10 LPM	0.5 HP	25 L	3/4" BSP
WCP 020	2.0 TR	7.0 kW	2.5 kW	230V / 1Ph / 50Hz	21 LPM	0.75 HP	40 L	1" BSP
WCP 030	3.0 TR	10.5 kW	3.4 kW	415V / 3Ph / 50Hz	32 LPM	1.0 HP	60 L	1" BSP
WCP 050	5.0 TR	17.5 kW	6.0 kW	415V / 3Ph / 50Hz	52 LPM	1.5 HP	80 L	1.25" BSP
WCP 075	7.5 TR	26.3 kW	8.5 kW	415V / 3Ph / 50Hz	78 LPM	2.0 HP	120 L	1.5" BSP
WCP 100	10.0 TR	35.1 kW	11.2 kW	415V / 3Ph / 50Hz	105 LPM	3.0 HP	160 L	2" BSP
WCP 150	15.0 TR	52.7 kW	16.5 kW	415V / 3Ph / 50Hz	158 LPM	3.0 HP	250 L	2" BSP
WCP 200	20.0 TR	70.3 kW	22.0 kW	415V / 3Ph / 50Hz	210 LPM	5.0 HP	350 L	2.5" BSP
WCP 300	30.0 TR	105.5 kW	33.0 kW	415V / 3Ph / 50Hz	315 LPM	7.5 HP	450 L	3" Flange
WCP 500	50.0 TR	175.8 kW	54.0 kW	415V / 3Ph / 50Hz	525 LPM	10.0 HP	700 L	4" Flange

ENGINEERING SELECTION & OPERATIONAL NOTES

- Dual refrigeration circuits provided standard on models WCP 150 and above for N+1 reliability.
- Sub-zero glycol chilling configurations (-15°C) available with 30–40% food-grade propylene or ethylene glycol brine.
- Custom water-cooled configurations available with shell-and-tube condensers for tower water integration.

FIELD DEPLOYMENT & COMPETITIVE ANALYSIS

Applications & Industrial Benchmarks

Demonstrated operational benefits across high-reliability manufacturing sectors

Plastic Injection Molding & Blow Extrusion
Cuts mold cycle times by up to 28% through rapid heat extraction and prevents part warping across multi-cavity tools.

Fiber Laser Cutting & Welding
Maintains resonator and cutting head optics at exact temperatures, preventing thermal blooming and beam focal drift.

CNC Electro-Spindles & Wire EDM
Extracts motorized spindle stator heat and stabilizes dielectric deionized water baths for sub-micron machining tolerances.

Chemical Reactors & Batch Jacketing
Controls aggressive exothermic reaction temperatures, preventing runaway chemical kinetics and product degradation.

Food, Beverage & Brewery Pasteurization
Cools wort fermentation tanks, jacketed dairy vessels, and soda carbonation carbonators to exact food safety specs.

Industrial Advantage: Win WCP vs. Commercial Comfort Air Chillers		Standard ISO 9001:2015 Quality
Engineering Parameter	Win Equipments Engineered Solution	Standard / Imported Benchmark
Hydronic Loop	Integrated SS304 buffer tank + SS pump in compact footprint	Requires external reservoir, external pumps, extra field piping
Thermal Shock	Oversized reservoir dampens instant 100% mold dump load	Microchannel freezing and frequent low-pressure trips
Ambient Rating	Heavy tropical coils rated for continuous 45°C ambient	Trips on high discharge pressure at 38°C ambient
Service Footprint	Direct factory field engineering team across South India	Third-party residential HVAC contractors with slow response

MANUFACTURING STANDARDS & FACTORY VERIFICATION

Factory Quality Assurance & Consultation

Calibrated testing regimes, global certifications, and direct engineering contact

✓ Factory Acceptance Testing (FAT)

- Calorimeter load cell testing across 10% to 110% simulated thermal capacity
- Helium mass-spectrometer refrigeration circuit leak detection (<0.1 g/year)
- 1.5× Hydrostatic pressure test on hydronic buffer tank and wetted piping
- Full 24-hour continuous burn-in test verifying PID temperature stability $\pm 0.5^{\circ}\text{C}$

★ Certified Standards & Accreditations

- ISO 9001:2015 Quality Management System
- IAF (International Accreditation Forum) Accredited
- DAC (Dubai Accreditation Center) Recognized
- Pre-dispatch run-testing under tropical temperature simulation



Manufacturer Warranty Assurance: 12 Months comprehensive manufacturer warranty on complete chiller skid and compressor. Direct field service engineering and genuine replacement spare parts stocked at Arasur Works, Coimbatore.

WIN EQUIPMENTS

SAVE WATER AND POWER • ISO 9001:2015 (IAF & DAC ACCREDITED)

📍 SF No: 4, 195 B, Kallangadu, Arasur Post, Coimbatore – 641407, Tamil Nadu, India

☎ +91 95972 28969 / +91 95972 28975

✉ info@winequipments.com • winequipments.com



SCAN FOR WEB
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